

Programme	B. Sc. Computer Science Minor				
Course Title	Web Design Trends and Techniques				
Type of Course	Minor				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Knowledge in Computers. 2. Basic knowledge in Internet and Basic knowledge Computers and Internet				
Course Summary	The aim of this course is to provide students with an understanding of the basic concepts in web browser and to achieve the basic web designing skills				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To get general introduction to internet	U	C	
CO2	To identify and analyse the current trends in web designing	Ap	P	
CO3	To understand basic knowledge in HTML5 and CSS3 for responsive web design	Ap	P	
CO4	To learn how to design a simple web applications	Ap	P	
CO5	To incorporate user experience principle in web design	Ap	P	
CO6	To Enable student to become	Ap	P	

	proficient in web designing through current technologies			
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs
I		Introduction to Web Design (15 marks)	09
	1	Overview of Internet	1
	2	Over view of Internet Security	1
	3	Client Server System	1
	5	Websites and Digital Communication Tools	1
	6	Collaboration for Website Development	1
	7	Understanding the evolution of web design	2
	8	Exploring current design trends	1
	9	Overview of innovative websites	1
II		HTML – Building the Foundation (18 Marks)	12
	10	Understanding the basic structure of web pages(Role of HTML, basic concept of webpage, html document structure <!DOCTYPE>,<html>,<head>,<body>).	2
	11	Exploring tags, attributes, and their significance (font type, text formatting tag, otrher text related tag, heading, paragraphs, list, link, image, common attributes like class, id, src, alt, href).	2
	12	Creating interactive forms to collect user data(form element tag like <form>,<input>,<textarea>,<select>,<button>,various form controls like text input, button, drodwonbox).	3
	13	Designing and structuring tabular data.(Basic table structure tag, colspan, rowspan)	2
	14	Enhancing the meaning and structure of your content(understanding semantic elements, benifit of semantic	3

		HTML).	
III	CSS – Styling Your Web Pages (18 Marks)		12
	15	Introduction to CSS(understanding the concept of styling, presentation, basic syntax selectors, properties and values, inline, internal and external style, different types of selectors)	3
	16	Introduction to common CSS properties(color,font,text,margin, padding, border, background), CSS box model(margin, border, padding and content), positioning elements (static, relative, absolute, fixed, z-index)	3
	17	Introduction to layouts in web designing, The role of responsive layouts, Flexbox layout model, creating Grid, Media queries and breakpoints	3
	18	Transition properties(duration, timing function, property), CSS Animation, Adding interactive hover effect. Overview of CSS frameworks and their benefits. Introduction to Bootstrap and its features.	3
IV	JavaScript Essentials (19 Marks)		12
	19	Overview of Javascript, declaring the variables and understanding data types. Object in Javascript, basic operations and control flow in Javascript	3
	20	Understanding the Document Object Model (DOM).Using selectors to target HTML elements. Modifying content, attributes, and styles dynamically. Using selector, content, attributes and styles dynamically. Creating and Deleting elements. Navigating through the DOM hierarchy.	3
	21	Understanding events triggered, common events, writing event handlers, Bubbling and capturing phases of event propagation, controlling event flow, accessing event object	3
	22	Overview of JavaScript libraries and their benefits.Selecting elements, manipulating the DOM, and handling events with jQuery.Applying visual effects and animations with jQuery. Fade, slide, show/hide, and custom animations.Making asynchronous requests with jQuery.ajax(). Handling JSON data and updating the DOM dynamically.	3

V	Hands-on Programming in Java(Using VSCode, Atom, Aptana Studio):		30
	Practical Applications, Case Study and Course Project		
1	Implement the following:		20
	1. Program for implementing html tags		
	2. Write program for implementing Style a paragraph (<p>) to have a red color, a font size of 16px, and a bold font weight		
	3. Write a program to implement CSS Box Model		
	4. Create a simple layout using Flexbox, with three div elements aligned horizontally.		
	5. Implement a media query that changes the background color of a webpage when the screen width is less than 600 pixels.		
	6. Write a JavaScript function that changes the text content of an HTML element with the id "demo" to "Hello, World!" when a button is clicked.		
	7. Declare a variable in JavaScript and assign it a string value. Also, mention the data type of the variable.		
	8. Write a JavaScript program for attaching a click event to a button.		
	9. Use JavaScript to make an asynchronous request to a JSON file and display the data on the webpage.		
	10. Use JavaScript to perform AJAX operation		
2	Case Study	2	
3	Project: Build a web application for perform responsive web application.	8	

Text Book :

1. HTML5 Black Book, Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP And JQuery (Second Edition), Dreamtech Press,ISBN: 9789351199076

References :

1. Internet and World Wide Web, H.M.Dietel, Pearson.
2. Mastering HTML, CSS & Javascript Web Publishing (English, Paperback, Lemay Laura)
3. Web Designing (English, Paperback, Hirdesh Bhardwaj)

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	3	3	-	-						
CO 2	1	-	3	3	-	-						
CO 3	-	-	3	3	2	3						
CO 4	-	-	2	3	-	-						
CO 5	-	-	3	3	2	3						
CO 6			3	3	3							

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓